Single-Set Quantitative Selection Principle

This note implements the concrete selection-principle step from `plan1.md`. The purpose is not to replace the whole Brasco–De Philippis–Velichkov proof, but to isolate the part that can be made quantitative without changing their free-boundary machinery.

The main output is a single-set form of BDV Proposition 4.4. Starting from one near-counterexample, it constructs a selected quasi-open minimizer whose energy-to-asymmetry ratio is controlled up to constants.

1 Normalization

Use the BDV torsion convention

$$E(\Omega) = \min_{u \in H_0^1(\Omega)} \left\{ \frac{1}{2} \int_\Omega |\nabla u|^2 - \int_\Omega u \right\} = -\frac{1}{2} \int_\Omega u_\Omega.$$

Fix

$$|\Omega| = |B_1| = \omega_N.$$

Then the Saint-Venant deficit is

$$\delta(\Omega) := E(\Omega) - E(B_1) \geq 0.$$

For the selection step, use BDV's smoother asymmetry

$$\alpha(\Omega) := \int_{\Omega \Delta B_1(x_\Omega)} \big|1 - |x - x_\Omega|\big| \, dx,$$

where x_Ω is the barycenter. BDV Lemma 4.2 gives, for bounded $\Omega \subset B_R$,

$$|\Omega \Delta B_1(x_\Omega)|^2 \lesssim_N \alpha(\Omega),$$

$$|\alpha(\Omega_1) - \alpha(\Omega_2)| \lesssim_R |\Omega_1 \Delta \Omega_2|,$$

and, for nearly spherical sets,

$$\alpha(\Omega) \lesssim_N \|\varphi\|^2_{L^2(\partial B_1)}.$$

Thus the right badness ratio for the selection step is

$$Q_\alpha(\Omega) := \frac{E(\Omega) - E(B_1)}{\alpha(\Omega)}.$$

The final Fraenkel statement is recovered later from $\mathcal{A}(\Omega)^2 \lesssim \alpha(\Omega)$.

2 Penalized Problem

Let $\Omega_0 \subset B_R$, $R \geq 2$, satisfy

$$|\Omega_0| = |B_1|, \qquad \varepsilon := \alpha(\Omega_0) > 0, \qquad \delta := E(\Omega_0) - E(B_1), \qquad q := \delta/\varepsilon.$$

Assume $q \ll 1$. Choose

$$\tau := q^{1/4},$$

or more generally any τ with

$$q \ll \tau^2 \ll 1.$$

Let $f_{\hat{\eta}}$ be the BDV volume penalty from Lemma 4.5:

$$f_{\hat{\eta}}(s) = \begin{cases} \hat{\eta}(s - \omega_N), & s \leq \omega_N, \\ (s - \omega_N)/\hat{\eta}, & s \geq \omega_N. \end{cases}$$

Set

$$F_{\hat{\eta}}(U) := E(U) + f_{\hat{\eta}}(|U|).$$

BDV Lemma 4.5 says that, for $\hat{\eta} = \hat{\eta}(N, R)$, B_1 is a minimizer of $F_{\hat{\eta}}$ among sets in B_R.

Now minimize over quasi-open $U \subset B_R$:

$$\mathcal{G}_{\varepsilon,\tau}(U) := F_{\hat{\eta}}(U) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2}.$$

This is exactly BDV's penalized functional (4.10), with the sequence parameter σ replaced by the single-set parameter τ.

Constants and Parameter Regime

The single-set proof uses the following constants from BDV Section 4: $\hat{\eta} = \hat{\eta}(N, R)$ and $C_4 = C_4(N, R)$ from Lemma 4.5; $C_2 = C_2(R)$, the Lipschitz constant of α on subsets of B_R from Lemma 4.2(ii); $\sigma_2(N, R)$ from Lemma 4.9; and $\sigma_3(N, R)$ from Lemma 4.16. Separate

$$\tau_{\text{ex}}(N, R) \leq \frac{\hat{\eta}}{2C_2}$$

from

$$\tau_{\text{reg}}(N, R) \leq \min\{\tau_{\text{ex}}(N, R), \sigma_2(N, R), \sigma_3(N, R)\}.$$

The first threshold is enough for existence and the uniform perimeter bound in the proof of BDV Lemma 4.6. The second is used when the Alt–Caffarelli regularity package is needed. All constants below may depend on N, R, and once $\tau \leq \tau_{\text{reg}}$ the free-boundary constants are uniform.

3 Quantitative Selected-Minimizer Estimates

Assume $\tau \leq \tau_{\text{ex}}(N, R)$, small enough that the BDV existence and perimeter estimates hold with τ in place of σ. Let U_* be a minimizer of $\mathcal{G}_{\varepsilon,\tau}$.

Lemma 1. Energy and Asymmetry Preservation Before Normalization

The minimizer U_* satisfies

$$0 \leq F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_1) \leq \delta,$$

and

$$|\alpha(U_*) - \varepsilon| \leq \frac{\sqrt{2\varepsilon\delta + \delta^2}}{\tau}.$$

In particular, if $q = \delta/\varepsilon \leq 1$ and $\tau = q^{1/4}$, then

$$|\alpha(U_*) - \alpha(\Omega_0)| \leq C\alpha(\Omega_0)q^{1/4}.$$

Proof. By minimality against Ω_0,

$$F_{\hat{\eta}}(U_*) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U_*) - \varepsilon)^2} \le F_{\hat{\eta}}(\Omega_0) + \varepsilon.$$

Since $|\Omega_0| = |B_1|$,

$$F_{\hat{\eta}}(\Omega_0) = E(\Omega_0) = F_{\hat{\eta}}(B_1) + \delta.$$

Because B_1 minimizes $F_{\hat{\eta}}$,

$$F_{\hat{\eta}}(U_*) \ge F_{\hat{\eta}}(B_1).$$

Therefore

$$F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_1) \le \delta,$$

and

$$\sqrt{\varepsilon^2 + \tau^2(\alpha(U_*) - \varepsilon)^2} \le \varepsilon + \delta.$$

Squaring the last inequality gives

$$\tau^2(\alpha(U_*) - \varepsilon)^2 \le 2\varepsilon\delta + \delta^2.$$

This proves the claim.

Lemma 2. Volume Error

There is $C = C(N, R)$ such that

$$\big||U_*| - |B_1|\big| \le C\delta.$$

Proof. Let B_{r_*} be a ball with $|B_{r_*}| = |U_*|$. By rearrangement,

$$F_{\hat{\eta}}(B_{r_*}) \le F_{\hat{\eta}}(U_*).$$

BDV Lemma 4.5 gives

$$F_{\hat{\eta}}(B_{r_*}) - F_{\hat{\eta}}(B_1) \ge \frac{|r_* - 1|}{C_4(N, R)}.$$

Together with Lemma 1,

$$|r_* - 1| \le C\delta.$$

Since $|B_{r_*}| = \omega_N r_*^N$, this implies

$$\big||U_*| - |B_1|\big| \le C\delta.$$

Lemma 3. Ratio Preservation After Volume Normalization

Let

$$\lambda := \left(\frac{|B_1|}{|U_*|}\right)^{1/N}, \qquad \widetilde{\Omega} := \lambda U_*.$$

After translating if needed, $\widetilde{\Omega}$ has volume $|B_1|$, and

$$\alpha(\widetilde{\Omega}) \ge \left(1 - Cq^{1/4}\right)\alpha(\Omega_0),$$

$$E(\widetilde{\Omega}) - E(B_1) \le C\big(E(\Omega_0) - E(B_1)\big),$$

provided $q \le q_0(N, R)$ and $\alpha(\Omega_0)$ is in the small regime. Consequently,

$$Q_\alpha(\widetilde{\Omega}) \le CQ_\alpha(\Omega_0).$$

Proof sketch. Lemma 2 gives

$$|\lambda - 1| \le C\delta = C\varepsilon q.$$

Since $q \ll 1$, this error is smaller than the asymmetry-preservation error $\varepsilon q^{1/4}$. The selected minimizer has the perimeter bound from the BDV existence argument, so the small dilation satisfies

$$|U_* \Delta \lambda U_*| \le C(N,R)|\lambda - 1|.$$

The Lipschitz property of α on bounded sets then gives

$$|\alpha(\widetilde{\Omega}) - \alpha(U_*)| \le C\varepsilon q.$$

Combining with Lemma 1 gives

$$|\alpha(\widetilde{\Omega}) - \varepsilon| \le C\varepsilon q^{1/4}.$$

For the energy, use

$$E(\lambda U_*) = \lambda^{N+2} E(U_*),$$

the bound $F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_1) \le \delta$, the Lipschitz control of $f_{\hat{\eta}}$ in the volume variable, and $|\lambda - 1| \le C\delta$. Since $U_* \subset B_R$, the energy is bounded in terms of N, R, so the dilation changes the deficit by $O_{N,R}(\delta)$. Thus

$$E(\widetilde{\Omega}) - E(B_1) \le C\delta.$$

The ratio estimate follows from the two displayed bounds.

Proposition 4. Single-Set Selection Map Preserving Q_α

Fix $R \ge 2$. There are constants

$$C_{\mathrm{sel}}, P_{\mathrm{sel}}, q_{\mathrm{sel}}, \varepsilon_{\mathrm{sel}} > 0$$

depending only on N, R with the following property. Let $\Omega_0 \subset B_R$ be open or quasi-open with

$$|\Omega_0| = |B_1|, \qquad 0 < \varepsilon := \alpha(\Omega_0) \le \varepsilon_{\mathrm{sel}}, \qquad q := Q_\alpha(\Omega_0) = \frac{E(\Omega_0) - E(B_1)}{\alpha(\Omega_0)} \le q_{\mathrm{sel}}.$$

Choose

$$\tau = q^{1/4},$$

with $q_{\mathrm{sel}} \le \tau_{\mathrm{ex}}^4$, minimize $\mathcal{G}_{\varepsilon,\tau}$, and volume-normalize the minimizer by

$$\widetilde{\Omega} = \left(\frac{|B_1|}{|U_*|}\right)^{1/N} U_*.$$

Then

$$P(U_*) \le P_{\mathrm{sel}}, \qquad |\widetilde{\Omega}| = |B_1|,$$

$$\alpha(\widetilde{\Omega}) \ge \left(1 - C_{\mathrm{sel}} q^{1/4}\right) \alpha(\Omega_0) \ge \frac{1}{2}\alpha(\Omega_0),$$

$$E(\widetilde{\Omega}) - E(B_1) \le C_{\mathrm{sel}}\big(E(\Omega_0) - E(B_1)\big),$$

and therefore

$$Q_\alpha(\widetilde{\Omega}) \le 2C_{\mathrm{sel}} Q_\alpha(\Omega_0).$$

More generally, if $\tau \le \tau_{\mathrm{ex}}$ is not tied to q, set

$$\rho(q,\tau) := \frac{\sqrt{2q + q^2}}{\tau} + C_{\mathrm{sel}} q.$$

Whenever $q \le 1$, $\varepsilon \le \varepsilon_{\mathrm{sel}}$, and $\rho(q,\tau) \le 1/2$, the same argument gives

$$\alpha(\widetilde{\Omega}) \ge (1-\rho(q,\tau))\alpha(\Omega_0), \qquad Q_\alpha(\widetilde{\Omega}) \le \frac{C_{\mathrm{sel}}}{1-\rho(q,\tau)} Q_\alpha(\Omega_0).$$

This is the precise single-set replacement for the limiting estimate BDV obtain in Proposition 4.4(iv). The smallness of $\varepsilon_{\mathrm{sel}}$ is only used after selection, to keep the dilation inside a fixed bounded class and to make the later Hausdorff/flatness threshold available; the quotient preservation itself is governed by $\rho(q,\tau)$.

4 Free-Boundary Regularity Inherited from BDV

The minimizer $U_* = \{u_* > 0\}$ satisfies the perturbed one-phase minimality inequality

$$\frac{1}{2}\int |\nabla u_*|^2 - \int u_* + f_{\hat{\eta}}(|\{u_*>0\}|) \le \frac{1}{2}\int |\nabla v|^2 - \int v + f_{\hat{\eta}}(|\{v>0\}|) + C_R\tau|\{u_*>0\}\Delta\{v>0\}|$$

for all $v \in H_0^1(B_R)$. This is BDV (4.19) with τ in place of σ.

Hence, for $\tau \le \tau_{\mathrm{reg}}(N,R)$, the Alt-Caffarelli estimates used in BDV apply:

- nondegeneracy of u_* near the free boundary,
- Lipschitz control,
- density estimates for both phases,
- reduced-boundary structure,
- smoothness once the free boundary is sufficiently flat.

The Euler-Lagrange Bernoulli coefficient has the same structure as BDV Lemma 4.16:

$$q_{u_*}(x)^2 = \Lambda + \frac{\tau^2(\alpha(U_*)-\varepsilon)}{\sqrt{\varepsilon^2+\tau^2(\alpha(U_*)-\varepsilon)^2}} \cdot \Psi_{U_*}(x),$$

where Ψ_{U_*} is the smooth α-variation term. The prefactor is bounded by τ, so q_{u_*} has uniform C^k bounds near ∂U_*, exactly as in BDV Lemma 4.16.

5 Quantitative Compactness Pipeline

BDV pass from L^1-convergence to a global smooth graph through compactness. For a quantitative version, separate that passage into the following three lemmas. Lemma 5 is essentially proved by the density estimate; Lemmas 6 and 7 are the remaining places where the constants in Alt–Caffarelli's flatness theorem must be tracked.

Lemma 5. α-to-Hausdorff

Assume $U \subset B_R$ is open, $x_U = 0$, and satisfies the two-sided density estimates

$$|U \cap B_r(x)| \ge cr^N, \qquad |B_r(x) \setminus U| \ge cr^N$$

for every $x \in \partial U$ and every $r \le r_0$. If, after centering,

$$|U \Delta B_1| \le M,$$

then

$$d_H(\partial U, \partial B_1) \le C_H(N,R,c,r_0)M^{1/N}.$$

Therefore, by BDV Lemma 4.2(i),

$$|U\Delta B_1(x_U)| \le C_1(N)^{1/2}\alpha(U)^{1/2},$$

and selected minimizers satisfying BDV Lemma 4.11 obey

$$d_H(\partial U, \partial B_1(x_U)) \le C_H(N,R)\alpha(U)^{1/(2N)}.$$

Proof sketch. If $x \in \partial U$ and $\mathrm{dist}(x, \partial B_1) = d$, then for $r \sim d$ the ball $B_r(x)$ lies either inside B_1 or outside B_1. The density estimate forces a chunk of size $\gtrsim r^N$ in $U\Delta B_1$. Thus $d^N \lesssim M$.

Conversely, if $y \in \partial B_1$ is distance d from ∂U, then $B_d(y)$ is either contained in U or disjoint from U. Either way, a fixed fraction of $B_d(y)$ lies in $U\Delta B_1$, so again $d^N \lesssim M$.

Lemma 6. Hausdorff-to-Flatness at One Alt–Caffarelli Scale

Let u be a selected minimizer with $\tau \le \tau_{\mathrm{reg}}$, translated so that $x_U = 0$. Let $\bar\mu, \bar\kappa, \gamma, C_{\mathrm{AC}}$ be the constants in BDV Theorem 4.18, depending on the uniform bounds for q_u, $\mathrm{Lip}\, q_u$, R, and N. Fix $0 < \mu \le \bar\mu/16$, and choose $\rho_\mu > 0$ so that

$$\rho_\mu \le \bar\kappa\mu^2$$

and, for every $\bar x \in \partial B_1$,

$$\partial B_1 \cap B_{5\rho_\mu}(\bar x) \subset \{x : |(x-\bar x)\cdot \nu_{\bar x}| \le \mu\rho_\mu\},$$

where $\nu_{\bar x}$ is the interior normal to B_1. There exists $h_F = h_F(N, R, \mu, \rho_\mu) > 0$, for instance $h_F \lesssim \mu\rho_\mu$, such that if

$$d_H(\partial U, \partial B_1) \le h_F,$$

then for every $\bar x \in \partial B_1$ one can find $x_0 \in \partial U \cap B_{h_F}(\bar x)$ such that u is of class $F(C\mu, 1, +\infty)$ in $B_{4\rho_\mu}(x_0)$, in the sense of BDV Definition 4.17.

The proof should use only the Hausdorff inclusion, the ball curvature estimate above, the distance bounds $u \simeq \mathrm{dist}(\cdot, \partial U)$ from BDV Lemma 4.11, and the uniform bounds $0 < c \le q_u \le C$ from BDV Lemma 4.12. This is the first remaining constant-tracking task.

Lemma 7. Flatness-to-Global Graph

Under the hypotheses of Lemma 6, if $C\mu \le \bar\mu$, BDV Theorem 4.18 gives local $C^{1,\gamma}$ representations of ∂U in every ball $B_{\rho_\mu}(x_0)$ produced above, with local norm at most $C_{\mathrm{AC}}\mu$. If in addition $C_{\mathrm{AC}}\mu$ is smaller than a fixed tubular-neighborhood slope threshold, then these local graphs patch through the radial projection to a single function $g : \partial B_1 \to \mathbb{R}$:

$$\partial U = \{(1+g(\theta))\theta : \theta \in \partial B_1\},$$

with

$$\|g\|_{L^\infty(\partial B_1)} \le C\, d_H(\partial U, \partial B_1), \qquad \|g\|_{C^{1,\gamma}(\partial B_1)} \le C(N,R)\mu.$$

If q_u has the C^k-bounds from BDV Lemma 4.16, the same patching gives uniform $C^{k+1,\gamma}$-bounds. Since every boundary point is covered by an Alt–Caffarelli graph, this also rules out free-boundary singularities in the small-α regime.

Corollary 8. Explicit Graph-Entry Threshold

For selected minimizers with $\tau \le \tau_{\mathrm{reg}}$, fix $\mu = \mu(N,R)$ small enough for Lemmas 6 and 7, then define

$$\alpha_{\mathrm{graph}}(N,R) := \left(\frac{h_F(N,R,\mu,\rho_\mu)}{C_H(N,R)}\right)^{2N}.$$

If

$$\alpha(U_*) \le \alpha_{\mathrm{graph}}(N,R),$$

then ∂U_* is a global $C^{1,\gamma}$ graph over $\partial B_1(x_{U_*})$. After the volume normalization in Lemma 3, the same is true for $\partial\widetilde{\Omega}$, with constants changed only by a factor controlled by N, R.

The quantitative route after sorting is therefore

$$\alpha(U_*) \le \alpha_{\mathrm{graph}} \Longrightarrow \text{Hausdorff closeness} \Longrightarrow \text{fixed-scale flatness} \Longrightarrow \text{global graph.}$$

Once the graph entry is available, BDV's nearly spherical second variation applies:

$$E(\widetilde{\Omega}) - E(B_1) \gtrsim_N \|\varphi\|^2_{H^{1/2}(\partial B_1)} \gtrsim_N \alpha(\widetilde{\Omega}),$$

contradicting $Q_\alpha(\widetilde{\Omega}) \ll 1$.

Thus the selection route has been reduced to two concrete missing quantitative regularity statements: prove Lemma 6 with explicit dependence on the density and distance constants, and record the patching constants in Lemma 7.